A

PROJECT REPORT

On

LokVeda — Your E-Gram Panchayat

Submitted In partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

Computer Science & Engineering

(Maulana Abul Kalam Azad University of Technology)

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Table Of Contents

Overview of Project

LokVeda is a web-based E-Gram Panchayat platform designed to digitize village-level administrative services and citizen interactions. The system aims to provide a secure, user-friendly, and transparent environment where citizens can apply for government services, track application status, and interact with Panchayat authorities digitally.

The project implements a role-based architecture consisting of Citizens, Staff Members, and Administrators. Each role is provided with specific privileges and workflows to ensure proper application processing and governance.

LokVeda focuses on improving accessibility, reducing paperwork, enhancing transparency, and streamlining administrative operations through secure authentication, location verification, and multi-stage application approval workflows.

Literature Review

Traditional village administration systems largely depend on paper-based workflows, physical visits to Panchayat offices, and manual record keeping. These approaches often result in:

- Delayed service delivery
- Data redundancy
- Lack of transparency
- Difficulty in application tracking
- Administrative inefficiencies

Several E-Governance initiatives introduced by the Government of India have demonstrated the effectiveness of digital platforms in improving public service delivery.

Research conducted on digital governance systems highlights the importance of:

- Secure user authentication
- Role-based access control
- Service digitization
- Workflow automation
- Citizen transparency

LokVeda incorporates these principles and adapts them for a Panchayat-level governance model.

System Block Diagram

```
+--------------------+
|      Citizen       |
+---------+----------+
          |
```

```
          v
+---------------------+
| Authentication &    |
| Location Validation|
+----------+----------+
           |
           v
+---------------------+
|   Service Portal    |
+----------+----------+
           |
           v
+---------------------+
| Application System |
+----------+----------+
           |
           v
+---------------------+
|      Staff          |
| Review Process      |
+----------+----------+
           |
           v
+---------------------+
|      Admin          |
| Approval Process    |
+----------+----------+
           |
           v
+---------------------+
| Approved Services   |
+---------------------+

           ^
           |
+---------------------+
| MongoDB Database    |
+---------------------+
```

Project Methodology

The project follows an iterative development methodology.

Phase 1: Requirement Analysis

- Studied existing E-Governance systems
- Identified user roles
- Defined service workflows
- Established security requirements

Phase 2: System Design

- Designed MVC architecture
- Defined database models
- Planned authentication workflow
- Designed user interfaces

Phase 3: Development

Implemented:

- Authentication module
- Session management
- Role-based dashboards
- Service management system
- Application workflow
- Status tracking system

Phase 4: Testing

Conducted:

- Authentication testing
- Authorization testing
- Form validation testing
- Workflow testing
- UI/UX testing
- Edge case testing

Phase 5: Deployment Preparation

- Documentation generation
- Security review
- GitHub repository preparation
- Cloud deployment planning

Technical Platform

Frontend

- HTML5
- CSS3
- JavaScript
- EJS Templates

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose ODM

Authentication & Security

- JWT
- Bcrypt
- Crypto
- Cookie-based Sessions

Communication

- Nodemailer

Development Tools

- VS Code
- Git
- GitHub

Deployment Platform

- Render (Planned)
- Cloudflare (Planned)

Expected Result

The system is expected to:

- Digitize Panchayat-level service delivery
- Reduce dependency on manual paperwork
- Improve service transparency
- Enable secure authentication
- Provide application tracking facilities
- Streamline approval workflows
- Improve citizen accessibility

The final platform offers a complete workflow from application submission to approval while maintaining security and usability.

Scope of Further Improvements

Future versions of LokVeda may include:

Document Management

- File uploads
- PDF generation

- Document verification

Notification System

- Email notifications
- SMS notifications
- Approval alerts

Advanced Search

- Service search
- Application search
- Citizen search

Analytics Dashboard

- Application statistics
- Service utilization reports
- Administrative insights

Multilingual Support

- English
- Hindi
- Bengali
- Regional languages

Cloud Enhancements

- CDN integration
- DDoS protection
- Performance optimization

Mobile Support

- Progressive Web Application (PWA)
- Dedicated mobile application

References

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7. JWT Documentation

8. Bcrypt Documentation
9. Render Documentation
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